Prelim Skills Exam	
Course Code: CPE 212	**Program:** Computer Engineering
Course Name: Automating Server Management	**Date Performed:** 29/01/2026
Section: CPE31S1	**Date Submitted:** 29/01/2026
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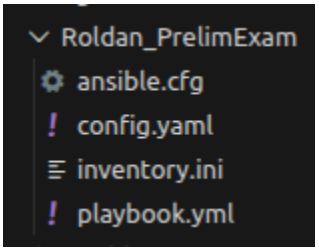
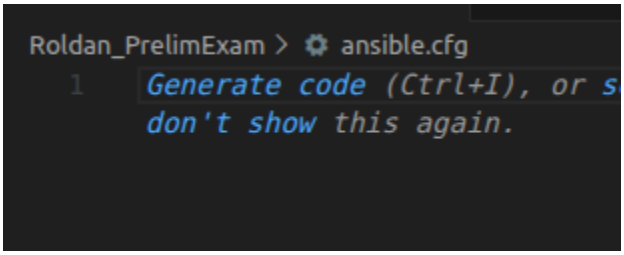
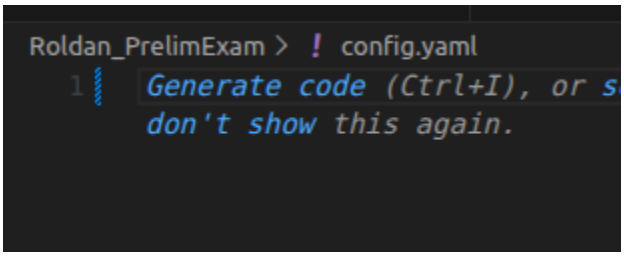


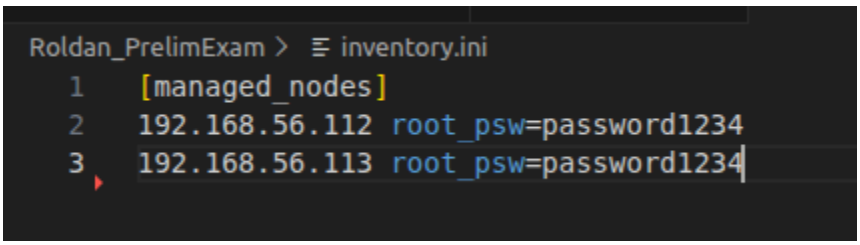
Final structure of the folder



The ansible.cfg is empty because no defaults were added



The config.yaml is empty because no variables were added



Inventory file contains IP addresses of the two managed nodes, as well as root passwords that were supposed to be used in setting up the root user login for mysql

```yaml
- name: Prelim Skills Exam
  hosts: all
  become: true

  vars:
    ansible_python_interpreter: /usr/bin/python3
    ansible_pip_interpreter: /usr/bin/python3-pip
    root_psw: password1234
```

Keyword 'name' sets the name of the play
Keyword 'hosts' determines the targeted machines. Since all machines for this exam are targeted, the parameter is set to 'all'
Keyword 'become' asks for the BECOME password before running the play

On the 'vars' section, 'ansible_python_interpreter' and 'ansible_pip_interpreter' set the default interpreter to be used.

```yaml
  tasks:
  - name: install python3 and pip3
    package:
      name:
        - python3
        - python3-pip
      state: latest
```

The task installs python3 and pip3's latest versions

```yaml
    - name: install java openjdk
      package:
        name:
          - openjdk-25-jre-headless
        state: latest

    - name: install mariadb
```

The above task installs java openjdk's latest version

```yaml
    - name: install mariadb
      package:
        name:
          - mariadb-server
        state: latest
```

The above task installs mariadb-server's latest version

```yaml
- name: MariaDB
  hosts: all
  gather_facts: true
  become: true
  vars:
    mysql_root_password: "Password1"
  tasks:
```

New play that installs mariadb, sql community libraries, python setup tools, and sets up root credentials.

```yaml
  tasks:
    - name: remove the mysql-community-libs package
      apt:
        name: mysql-community-libs
        state: absent
    - name: remove the mysql-community-common
      apt:
        name: mysql-community-common
        state: absent
    - name: install mariadb
      apt:
        name:
          - mariadb-server
        state: latest
```

Code above removes the mysql community libraries and replaces it with the mysql community common package, also installs mariadb again

```yaml
    - name: install python-setuptools
      apt:
        name:
          - python-setuptools
        state: latest
    - name: install pip
      apt:
        name:
          - python3-pip
        state: latest
    - name: eh
      ansible.builtin.pip:
        name: pymysql
        state: present
```

Code above installs python3, python, and pymysql

```
        state: present
    - name: start mariadb
      service:
        name: mariadb
        enabled: true
        state: started
    - name: Setting up root credentials
      mysql_user:
          name: root
          login_user: root
          password: "{{ root_psw }}"
          login_host: "localhost"
          state: present
```

Code above sets up root user login (doesn't work due to missing credentials)

Output/Verification of installation:

https://github.com/Seijuro2004/Roldan_PrelimExam.git

```
serge@node1:~$ java --version
openjdk 25.0.1 2025-10-21
OpenJDK Runtime Environment (build 25.0.1+8-Ubuntu-122.04)
OpenJDK 64-Bit Server VM (build 25.0.1+8-Ubuntu-122.04, mixed mode, sharing)
```

Status of Java on first managed node

```
serge@node1:~$ mariadb --version
mariadb  Ver 15.1 Distrib 10.6.22-MariaDB, for debian-linux-gnu (x86_64) using
EditLine wrapper
serge@node1:~$ pip --version
pip 22.0.2 from /usr/lib/python3/dist-packages/pip (python 3.10)
```

Status of pip and mariadb on first managed node

```
serge@node1:~$ python3 --version
Python 3.10.12
```

Status of python on first managed node

```
serge@node2:~$ java --version
openjdk 25.0.1 2025-10-21
OpenJDK Runtime Environment (build 25.0.1+8-Ubuntu-122.04)
OpenJDK 64-Bit Server VM (build 25.0.1+8-Ubuntu-122.04, mixed mode, sharing)
serge@node2:~$ mariadb --version
mariadb  Ver 15.1 Distrib 10.6.22-MariaDB, for debian-linux-gnu (x86_64) using
EditLine wrapper
serge@node2:~$ pip --version
pip 22.0.2 from /usr/lib/python3/dist-packages/pip (python 3.10)
serge@node2:~$ python3 --version
Python 3.10.12
serge@node2:~$
```

Status of java, mariadb, pip, and python3 on second managed node

https://github.com/Seijuro2004/Roldan_PrelimExam.git